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Crossing Spatial and Temporal Boundaries in Globally Distributed Projects: A Relational Model of Coordination Delay

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In globally distributed projects, members have to deal with spatial boundaries (different cities) and temporal boundaries (different work hours) because other members are often in cities within and across time zones. For pairs of members with spatial boundaries and no temporal boundaries (those in different cities with overlapping work hours), synchronous communication technologies such as the telephone, instant messaging (IM), and Web conferencing provide a means for real-time interaction. However, for pairs of members with spatial and temporal boundaries (those in different cities with nonoverlapping work hours), asynchronous communication technologies, such as e-mail, provide a way to interact intermittently. Using survey data from 675 project members (representing 5,674 pairs of members) across 108 projects in a multinational semiconductor firm, we develop and empirically test a relational model of coordination delay. In our model, the likelihood of delay for pairs of members is a function of the spatial and temporal boundaries that separate them, as well as the communication technologies they use to coordinate their work. As expected, greater use of synchronous web conferencing reduces coordination delay for pairs of members in different cities with overlapping work hours relative to pairs of members with nonoverlapping work hours. Unexpectedly, greater use of asynchronous e-mail does not reduce coordination delay for pairs of members in different cities with nonoverlapping work hours, but rather reduces coordination delay for those with overlapping work hours. We discuss the implications of our findings that temporal boundaries are more difficult to cross with communication technologies than spatial boundaries.

Key words: virtual teams; computer-mediated communication and collaboration
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> I think that the impact of a [globally distributed] project that isn't well coordinated is slowness in response time or miscommunications. You get out of sync. We have a long turn-around between a question or an answer sent by one side of the ocean back and forth.—interview quote from a globally distributed project member.

Introduction

A wide range of terms—including distant, proximate, dispersed, collocated, and virtual—evoke the spatial boundaries inherent in distributed work. Global software development, information technology (IT) offshoring, and worldwide just-in-time manufacturing are a few of the business practices that rely on employees in different geographic locations (Aspray et al. 2006). Spatial boundaries, defined in terms of the geographic locations that physically separate people, are fundamental to the study of distributed work (Hinds and Kiesler 2002). Prior geographic proximity research has linked spatial boundaries (e.g., being located on a different floor rather than on the same floor) with a reduction in work outcomes such as task communication (Allen 1977), collaboration likelihood (Kraut et al. 1990), and teamwork quality (Hoegl and Proserpio 2004). Previous studies that examined geographic distance have also shown that spatial boundaries (e.g., different city or different country) decrease the likelihood of sharing knowledge among employees (Hansen and Lovas 2004, Tsai 2002).

As work continues to become more globally distributed across different cities and countries, spatial boundaries will undoubtedly overlap with temporal boundaries (Espinosa et al. 2003). Temporal boundaries, conceptualized as nonoverlapping work hours (9:00 A.M. to 5:00 P.M. Pacific Standard Time versus 9:00 A.M. to 5:00 P.M. Greenwich Mean Time) have the potential to be as disruptive as spatial boundaries. Unfortunately, empirical research has not kept up with theorizing and case studies about the temporal boundaries apparent in distributed work (Carmel 2006, Saunders et al. 2004, Watson-Manheim et al. 2002). In fact, a recent review of the literature on geographically dispersed teams found *no* examples of field studies that conceptually and empirically distinguished the impact of spatial dispersion from the impact of temporal dispersion (O'Leary and Cummings 2007). Our study attempts to fill this gap.

To illustrate the distinction between spatial boundaries and temporal boundaries, consider a software development project with members split between sites in Northern and Southern California. Though members on the project reside in different geographic locations, they share overlapping hours in a workday. Thus, if members encounter an urgent problem, or need to coordinate their work in real time, they have access to synchronous communication technologies (e.g., telephone, instant messaging (IM), or web conferencing) throughout the workday. And if they prefer to use an asynchronous communication technology like e-mail, they have a choice. Now, consider another software development project with members split between sites in Northern California and India. Because of the 13.5-hour time difference between sites (West–East), the window for synchronous interaction is severely limited and their choice of communication technologies is reduced. More important, members could experience a delay in coordinating their work if they rely on irregular or ineffective use of asynchronous communication. In both software development projects, the members encounter spatial boundaries. However, in the first project, there are no temporal boundaries, while in the second project, there are.

We are interested in the impact of spatial and temporal boundaries on *coordination delay*, rather than generic forms of delay, because we are interested in the coordinated action of pairs of project members. Many of the coordination problems associated with lack of co-presence because of spatial boundaries can be solved to some extent with communication technologies. In contrast, temporal boundaries change the synchronicity and timing of activities, so a decrease in overlapping work hours increases the likelihood of longer response and waiting periods, thus reducing flexibility and agility in this work context. Moreover, temporal boundaries narrow the window of opportunity for synchronous interaction between members, thus it can be a source of coordination delay. Therefore we posit that the effects of spatial and temporal boundaries on coordination delay can be distinguished by focusing on each boundary separately. We further suggest that understanding the role of coordination delay is critical to understanding how temporal and spatial boundaries affect project performance.

Drawing from coordination theory, we develop a relational model of coordination delay that takes into account spatial and temporal boundaries among pairs of project members, along with the communication technologies used to cross these boundaries. We then use this model to test hypotheses about the main effects of spatial boundaries, temporal boundaries, and communication technology on coordination delay, and the corresponding interaction effects between each boundary and communication technology use.

A Relational Model of Coordination Delay

Coordination has long been considered an important aspect of joint work, because people have to integrate their effort toward a common goal (Thompson 1967, Van de Ven et al. 1976). Coordination theory is based on the notion that people need to manage dependencies among activities through processes such as goal selection and task decomposition (Malone and Crowston 1994). In the software development projects described above, members selected which components to work on first as well as how to divide up work on the components. It follows that effective coordination would occur if components were selected in a way that minimized disruption to the sequence of development activities (e.g., core architecture developed before noncore features) and work was divided up in a manner that decreased the dependencies among subtasks (e.g., graphical user interface developed separately from database structure).

However, these types of development projects are inherently volatile and complex (Banker and Slaughter 2000), so it is often difficult to anticipate the sequencing and division of task activities. As a result, there are likely to be coordination delays, or time spent waiting for tasks to be processed by others (Malone 1987). Coordination delays are specific to the coordination activities among project members, rather than general task delays that can also include prioritization decisions, resource constraints, and supplier holdups. These time lags in coordination can be costly to organizations because of the additional hours worked by project members, and can be especially costly if more members are added to the projects (Brooks 1995). For members working on projects that span different geographic locations, coordination delay is an even greater concern, as it can take longer to receive information or decisions from members about the work (Herbsleb et al. 2000). Similarly, time zone differences can lead to delays in managing and exchanging project artifacts, such as updating customer requirements, product documentation, or progress reports, which hinders flexibility and agility in these knowledge-intensive environments (Agerfalk and Fitzgerald 2006). Given that project members can extend their work hours, shift their work hours, or even stay connected all day from home, work, or on the road, we do not know the extent to which coordination delay is caused by spatial or temporal boundaries, or both.

To capture the aspects of delay that involve coordination between any two members of a project, we introduce a relational model of coordination delay. Consistent with other relational models (e.g., Borgatti and Cross 2003, Granovetter 1973), we are interested in processes that occur within a relationship between any two members of a project. In terms of coordination delay, these processes include the additional time it takes for one person to resolve task issues, clarify task-related communication, and rework tasks with another person. Our relational model is different than team models of coordination, which aggregate dependencies and communication among all members to the project level of analysis (Van de Ven et al. 1976, Faraj and Sproull 2000). We believe that focusing on pairs of project members is critical for four reasons: (1) spatial and temporal boundaries are dyadic in nature, such that space or time separates one member from another, (2) factors such as coordination delay are both dyadic and directional, given that a project member may be the source of delay for one person and the recipient of delay from another, (3) aggregation to the project level can obscure coordination delays that occur between some pairs of members but not other pairs, and (4) a relational model gives us greater precision in connecting particular boundaries with coordination delay within a pair of members. Thus we extend coordination theory by taking a relational perspective on the link between spatial and temporal boundaries and coordination delay.

Spatial and Temporal Boundaries

Boundaries are "real or imagined lines that denote an edge or limit of something" (Cambridge Dictionaries Online, dictionary.cambridge.org). Thus spatial boundaries mark a threshold or cutoff because of geographic location, while temporal boundaries mark a threshold or cutoff because of overlapping work hours. In contrast to spatial or temporal distance, which are continuous constructs (e.g., 0 to 24,000 miles away or 0 to 23 time zones apart), spatial and temporal boundaries are discrete constructs (e.g., different cities or different work hours). For members of geographically dispersed projects, spatial and temporal boundaries often generate increased coordination costs. In the case of spatial boundaries, being in a different city often means that automobile or airplane travel is required to get from one city to the next, whereas being in a different building in the same city could mean walking or driving, depending on how close the buildings are to one another. Technology can be used to bridge both of these spatial boundaries, but being in different cities can also create additional coordination costs for members.

In the case of temporal boundaries, working in a different time zone often means that the amount of overlapping time in a day is reduced. For example, working 5 time zones apart from other members means that a few hours of overlap are available in a typical workday, whereas working 10 time zones apart from others means that no hours of overlap are available in a typical workday. Therefore the coordination costs are higher for pairs of members who do not have any overlap in a workday because they must shift work schedules, hold meetings during overlapping time, or communicate from home if they want to be synchronous. Again, technology can be used to bridge both of these temporal boundaries, but having some overlapping work hours in a day means that asynchronous and synchronous communication can be easily used, whereas having no overlapping work hours in a day means that only asynchronous communication can be easily used. In short, to cross the spatial boundary of a different city, members need to make travel arrangements and to cross the temporal boundary of different work hours members need to adjust when they work during the day. However, note that there are marginally diminishing coordination costs to being in a different city a little farther away (which still requires travel) and being a few more time zones apart (which still requires a change in schedule).

Prior work has shown that spatial boundaries can negatively impact the ease with which joint work is accomplished. For example, spatial boundaries reduce the likelihood of face-to-face contact and spontaneous communication, which are important for resolving problems that arise unexpectedly or informally discussing work issues (Allen 1977, Kiesler and Cummings 2002, Olson and Olson 2000). Furthermore, spatial boundaries restrict the task context and common ground shared by pairs of members, which are necessary for creating mutual knowledge about who is doing what with whom (Clark and Brennan 1991, Cramton 2001). In addition, project members working together across spatial boundaries must rely on different forms of communication technology, which can limit the breadth and richness of interaction (Daft and Lengel 1986, Sproull and Kiesler 1991, Williams 1977). Taken together, the negative consequences for the frequency, timeliness, and quality of coordination among project members suggest that coordination delays are more likely to occur across spatial boundaries relative to no spatial boundaries. Initial findings by Herbsleb, Grinter, and others (e.g., Herbsleb et al. 2000, Grinter et al. 1999) support the association between geography and coordination delay, though like other studies, they do not tease apart the impact of spatial boundaries from temporal boundaries. Thus, controlling for temporal boundaries, we propose that spatial boundaries do indeed affect coordination delay.

Hypothesis 1 (H1). *Crossing spatial boundaries* (*i.e., same city versus different city with overlapping work hours*) *will be associated with an increase in coordination delay.*

Previous research has found that temporal boundaries can also negatively impact the flow of collaborative work. When temporal boundaries exist among pairs of members, it is more difficult to engage in synchronous communication, real-time problem solving, and concurrent workflow (Carmel 2006, Espinosa and Carmel 2004). Put differently, communication across temporal boundaries, such as nonoverlapping work hours, will be largely asynchronous, and can lead to longer response and issue resolution times because members are not in sync with one another (Saunders et al. 2004, O'Leary and Cummings 2007). Prior theory and research have illustrated how temporal coordination is important for activities such as synchronizing, scheduling, and allocating time for working together (Bardram 2000, Massey et al. 2003, McGrath 1990). It follows that coordinating across time zones should increase the likelihood of delays among project members because of difficulties in temporal coordination. We propose that, even after controlling for spatial boundaries, coordination delay should increase with temporal boundaries because of a decrease in overlapping work hours among pairs of members.

Hypothesis 2 (H2). *Crossing temporal boundaries* (*i.e., same work hours versus different work hours in different cities*) *will be associated with an increase in coordination delay.*

Communication Technologies

Practitioners and academics alike have been optimistic that various communication technologies will be able to help project members overcome distance to coordinate effectively (Gibson and Cohen 2003, Lipnack and Stamps 1997, Carmel 1999). Although there are documented examples of software development projects that successfully "follow the sun" and hardware development projects that do an excellent job of designing in the West (United States, Europe) and producing in the East (India, China), it is unclear if these are exceptions or the norm (Sarker and Sarker 2009).

Communication technologies allow project members to communicate at a distance through the use of audio, video, text, graphics, and other features. Researchers have categorized communication technologies according to whether they are used synchronously or asynchronously and in the same place or different places (e.g., Bullen and Bennett 1993, McGrath and Hollingshead 1994). For example, telephone and e-mail are often used when two people are in different places, but telephone communication is synchronous while e-mail communication is asynchronous (in its dominant use). In projects separated by spatial boundaries such as different cities, face-to-face communication is not a regular option, given that members are not in the same place. Therefore, members are less likely to bump into one another in the hallway, see each other in the lunchroom, or encounter one another in meetings throughout a workday. As mentioned above, the result is that members will have less mutual knowledge about project members in other locations, including contextual information such as work schedules, time commitments, and other task constraints (Cramton 2001).

When project members are in different geographic locations, opportunities to update one another on progress and develop mutual knowledge through synchronous communication are more difficult to create. Furthermore, synchronous communication is less likely to occur spontaneously when project members are spread across spatial boundaries given the need to be available at the same time (Sarbough-Thompson and Feldman 1998). However, when project members in different geographic locations have time overlap in their workday, synchronous communication should reduce the likelihood of coordination delay because they can interact in real time to resolve issues quickly. Furthermore, relative to pairs of members that cross temporal boundaries (i.e., no time overlap in their workday), pairs of members who do not cross temporal boundaries have more synchronous time available for interaction and can make more effective use of synchronous communication, thus should be more effective at reducing coordination delay through synchronous communication. We hypothesize that

Hypothesis 3 (H3). *An increase in synchronous communication will be associated with a reduction in coordination delay.*

Hypothesis 4 (H4). *An increase in synchronous communication will be more strongly associated with a reduction in coordination delay than when members cross spatial boundaries.*

Hypothesis 5 (H5). *When members cross spatial boundaries, an increase in synchronous communication will be more strongly associated with a reduction in coordination delay than when members do not cross temporal boundaries.*

For project members separated by spatial and temporal boundaries, such as project members located on different sides of the globe, synchronous communication is even less likely to happen by chance. Given the restriction in which technologies are available to use when members do not have overlapping hours in a workday, one method for members to mitigate this challenge is through asynchronous communication such as e-mail. Research suggests that managers prefer e-mail for a wide assortment of activities, and that e-mail it can be used to share information, coordinate work, and construct a shared identity for the group (Finholt and Sproull 1990, Markus 1994). Through asynchronous communication, members can create awareness of what activities they are working on in the project and can solicit feedback from others even when they are not available. This awareness and feedback seeking is critical, especially when two members need to schedule time to communicate synchronously (e.g., an early morning or a late-night phone call) (Dourish and Bly 1992). Therefore we posit that more use of asynchronous communication tools will be associated with lower levels of coordination delay.

We also anticipate that this reduction in coordination delay will be stronger with temporal boundaries than without them because members who cross temporal boundaries will likely increase their use of e-mail relative to other communication technologies, which will translate into less coordination delay. Because of the advantage of using asynchronous communication for project members who do not work at the same time, we expect the following:

Hypothesis 6 (H6). *An increase in asynchronous communication will be associated with a reduction in coordination delay.*

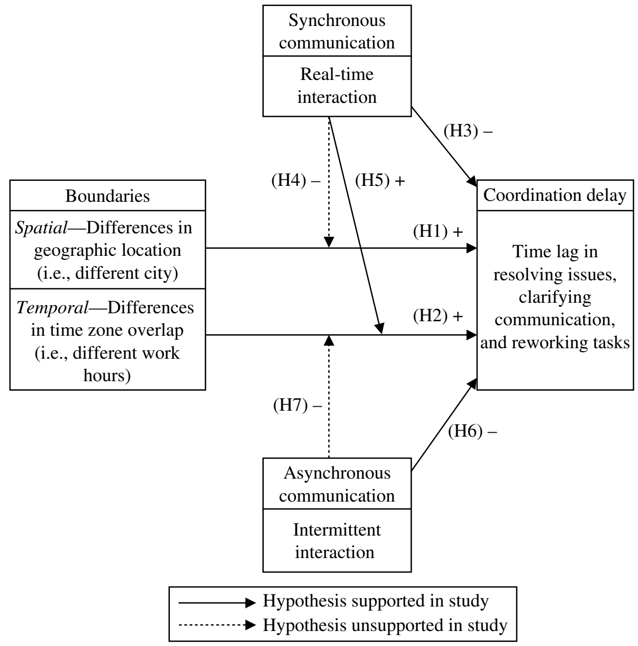
Hypothesis 7 (H7). *When members cross spatial boundaries, an increase in asynchronous communication will be more strongly associated with a reduction in coordination delay when members also cross temporal boundaries.*

Analysis Strategy

We test the above hypotheses using survey data from 675 members (representing 5,674 pairs of members) across 108 globally distributed projects in a *Fortune* 500 corporation. Each project had at least one member who worked in a different city than the other members, and more than half of the projects had at least one member who worked in a location where there was no overlap in a workday (i.e., greater than nine-hour time zone difference). Our relational model of coordination delay is displayed in Figure 1, and summarizes the linkages between spatial and temporal boundaries, communication technology use, and coordination delay. As expected, the distribution of member pairs reinforced the overlap between spatial and temporal boundaries (i.e., 30% same city and same work hours in a day, 43% different city and same work hours in a day, and 27% different city and different work hours in a day).

Given the co-occurrence of spatial and temporal boundaries in the case of project members spread across the world, we highlight our strategy for analyzing coordination delay. For hypotheses focused on the main effects of spatial boundaries (H1) and synchronous communication (H3) on coordination delay, as well as the interaction effect of spatial boundaries, and synchronous communication (H4), we hold temporal boundaries constant by examining $N = 4,165$ pairs of project members who have overlapping work hours. For hypotheses that address the main effects of temporal boundaries (H2) and asynchronous communication (H6) on coordination delay, as well as the interaction effects of temporal boundaries and synchronous communication (H5), and temporal boundaries and asynchronous communication (H7), we hold spatial boundaries constant by only looking at $N = 3,944$ pairs of project members who are in different cities but vary in whether they have overlapping work hours in a day.

Figure 1 A Relational Model of Coordination Delay in Globally Distributed Projects



Note. Statistically significant results in opposite direction.

We use hierarchical linear modeling (HLM) to analyze pairs of project members, given that some variables are measured at the level of the member (e.g., number of months worked on project, percentage of effort) and other variables are measured at the level of the relationship (e.g., communication technology use, coordination delay). HLM takes into account the nonindependence of observations, and adjusts the degrees of freedom to account for relationships nested within members, and members nested within projects (see Singer and Willett 2003 or Bryk and Raudenbush 1992 for a detailed discussion of multilevel models). For the subsample of pairs with overlapping work hours in a day, a NULL three-level model with no predictors of coordination delay showed that 1% of variance was explained between projects (level-3: intercept = 0.01, $p > 0.20$), 55% of variance was explained between members within projects (level-2: intercept = 0.44, $p < 0.01$), and 44% of variance was explained by relationships members had with others on the project (level-1: residual = 0.35, $p < 0.01$). For the subsample of pairs in different cities, a NULL three-level model with no predictors of coordination delay showed that 1% of variance was explained between projects (level-3: intercept = 0.01, $p > 0.20$), 61% of variance was explained between members within projects (level-2: intercept = 0.51, $p < 0.01$), and 38% of variance was explained by relationships members had with others on the project (level-1: residual = 0.32, $p < 0.01$). Given that there was no significant between-project variance at level 3 for either subsample, we conduct all analyses using a two-level model, where the level-1 predictors are "repeated measures" within members (Singer 1998), thus accounting for multiple observations (i.e., relationships) originating from the same member.

For all analyses, coordination delay is the dependent variable. Spatial boundaries (H1), temporal boundaries (H2), synchronous communication (H3), and asynchronous communication (H6) are the independent variables in the main effect models and then synchronous communication (H4 and H5) and asynchronous communication (H7) are the moderating variables in the interaction effect models. We center independent variables by the grand mean of all pairs of members, which is our theoretical focus given the relational model of coordination delay, so that the intercepts reflect the expected value of coordination delay for the "average" dyad across the sample (Hofmann and Gavin 1998). We note, however, that other forms of centering (for example, project mean) do not change the overall pattern of results.

Method

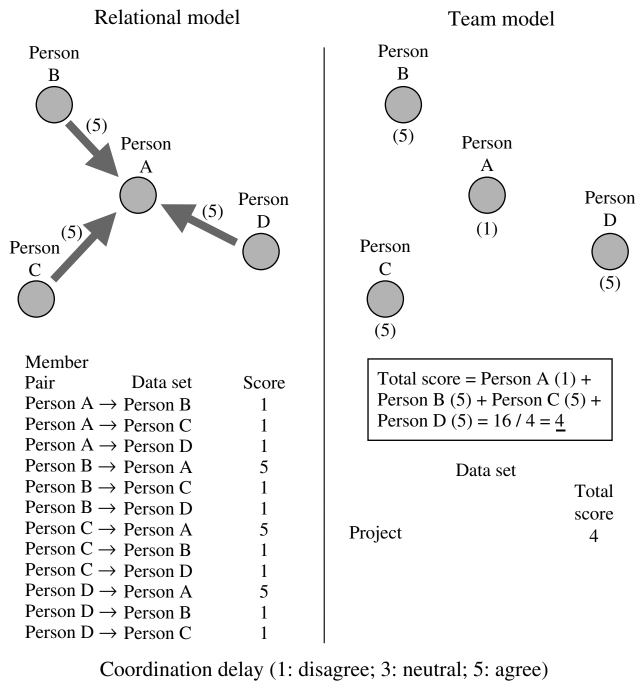
Sample

Participants from a large semiconductor manufacturing firm were solicited to participate in a study of global project effectiveness. Initially, we conducted semistructured interviews with 23 technical project members based in a number of geographic locations (e.g., United States, England, Ireland, Philippines, Israel, Russia, and China). These interviews provided information about the variety of technical work (e.g., software development, hardware development, systems integration), as well as the communication technologies used by globally distributed projects in the organization (e.g., e-mail, telephone, IM, web conferencing). As a next step, we randomly sampled roughly 4,000 managers from three large business units that rely heavily on technical projects. Project managers were asked to provide the name of a project they led in the prior six months along with the project start/end date and project description. Of the managers sampled, 380 provided project information. Given that the sample was representative of the population of project managers in the three business units (in terms of variety of technical work), we had no reason to believe that the sample was biased in any meaningful way.

Based on our interviews and pilot testing with several employees in the company, we felt that it was important to capture the extent to which spatial and boundaries impacted coordination delay for pairs of members. The data collection process required to test a relational model of coordination delay is different, and significantly more time intensive for the respondent, than the process required to test a team model of coordination delay (see Figure 2 for an example). We therefore decided to use an online survey to collect data on pairs of members in a project. Online surveys carry a range of benefits (quantifiable data, efficient data collection) and costs (self-reported responses, potential ambiguity of question wording) (Dillman 2000), though we felt that the benefits outweighed the costs. In addition to making minor wording changes based on feedback from the pilot testing to reduce question ambiguity, we took comfort in research demonstrating that respondents are relatively accurate in reporting long-term patterns of events compared with particular events in time (Freeman et al. 1987). Thus we ensured that our survey questions addressed patterns of interaction by asking respondents about the past six months of work on their projects.

The project managers who provided project information were requested to complete an online survey that included asking them to add the names of other people on the project. The online survey was dynamic,

Figure 2 An Example Distinguishing Between a Relational Model of Coordination Delay and a Team Model of Coordination Delay



Notes. In the above diagram, coordination delay is assessed in a four-person project. For the Relational model, an arrow points toward a source of coordination delay. Persons B, C, and D all report that Person A is a source of coordination delay (i.e., 5: agree). Note that arrows are not displayed when members indicate 1: disagree. For the Team model, members report the extent to which there is coordination delay in the project, but do not report the source of delay. Persons B, C, and D indicate 5: agree, while Person A indicates 1: disagree. In a data set for the Relational model, each pair of members is entered on a single row with the score for coordination delay, while in a data set for the Team model, the project is entered on a single row with the total score for coordination delay.

so once the project manager added other members, they were automatically sent an e-mail message inviting them to participate in the study and complete the survey. Of the managers providing project information, 300 of them provided the names of other people on the project. Across projects, a total of 2,318 names were generated, and 1,311 of them completed the survey, for an overall response rate of 57%. We did not find systematic differences between survey respondents and nonrespondents in terms of geographic location or role on project. Out of the completed responses, we distinguish between the 1,039 responses from project managers or project members (the "core" members) and the 272 responses from outside experts, stakeholders, or others affiliates (the "noncore" members). For purposes of our analyses, we only examine data from the core members. We reduce the sample further by only including data from 675 respondents (representing 5,674 pairs of core members) who were on a project with at least one other core member who responded. This ensures that we have at least two core member assessments of coordination delay and communication technology use in each project (see the appendix for an illustration of the data collection method).

Respondents in the final sample worked in 54 locations across 22 countries (Belgium, Brazil, Canada, China, Costa Rica, Denmark, France, Germany, India, Ireland, Israel, Japan, Malaysia, Mexico, Philippines, Poland, Russia, Singapore, South Korea, Taiwan, United Kingdom, and United States). More than half of the respondents were from engineering or IT, and worked on software or hardware projects. The typical project length was a little more than a year. Around 70% of the respondents were male, and the average age was 38. Respondents had, on average, over 10 years of industry experience and about 5 years of experience in the company.

Variables Measured at the Individual Level (i.e., Level-2: Project Members)

Project Manager. We included a binary variable for whether the respondent was the project manager (0: no, 1: yes). The manager often has a greater stake in the outcomes of a project, thus we wanted to control for the influence that being a project manager might have on coordination delay.

Number of Members. We counted the number of other core members on the project (i.e., project manager and project members) for which the respondent provided data. Because the online survey allowed respondents to add core members, some respondents could answer questions about coordination delay for more members than another respondent, so we control for number of members.

Number of Months. Members were asked to report the number of months that they personally worked on the project, and because members who are on a project longer have more opportunities to be affected by coordination delay, we control for number of months.

Percentage of Effort. To account for core members who devote more effort to their projects, and thus are more likely to be affected by coordination delay, we asked respondents to indicate the percentage of time they spent contributing to their project (1: 1%–10% of average work week, 2 = 11%–20% of average work week, 3 = 21%–30% of average work week, 4 = 31%–40% of average work week, 5 = 41%–50% of average work week, 6 = 51%–60% of average work week, 7 = 61%–70% of average work week, 8 = 71%–80% of average work week, 9 = 81%–90% of average work week, 10: 91%–100% of average work week).

Time Shifting. For the roughly 20% of project members reporting workday hours outside of 7:00 A.M. to 7:00 P.M. local time, and thus could reduce coordination delay, we created a binary variable to account for possible shifting of their work time (0: no time shifting, 1: time shifting). We conducted sensitivity analyses (e.g., using continuous and categorical measures) to ensure that our results did not change when using different measures of time shifting instead.

Variables Measured at the Relational Level (i.e., Level-1: Pairs of Project Members)

Years Known. An important control variable in any relational model, the respondent reported the number of years knowing each core member on the project (1: <than 1 year; 2: 1–3 years; 3: 3–5 years; 4: 5–10 years; 5: more than 10 years).

Member Interdependence. To ensure that we capture required coordination among core members, we use the average of a three-item scale measuring the extent to which core members depended on one another (i.e., coordination dependency occurs when members rely on each other to complete tasks, i.e., when one member depended on another member for information or materials. Please mark how often, during the past six months… (1) tasks this person performed were related to tasks I performed, (2) this person depended on me for information or materials needed to complete his or her work, (3) I could not accomplish my tasks without information or materials from this person; 1: not at all; 3: sometimes; 5: very much; Cronbach's $\alpha = 0.90$) (adapted from questions written at the project level by Campion et al. 1993).

Coordination Delay. We used a three-item scale measuring the extent to which there were delays in coordination among core members (i.e., these questions refer to how long you had to wait for information, materials, or task completion, to continue your project work during the past six months… (1) typically, it took a long time to get a response from this person, (2) our communications required frequent clarification, (3) we often had to rework tasks beyond what I would normally expect; 1: disagree; 3: neutral; 5: agree; Cronbach's $\alpha = 0.79$) (derived from Espinosa and Carmel 2004).

Spatial Boundaries. Company database records were used to determine the city location of each project member. Given that almost all cities had a single company site, we used city as the cutoff for the binary measure of spatial boundaries (0: same city, 1: different city).

Temporal Boundaries. More than 80% of the sample reported working between 9–11 hours a day, arriving 7:00 to 9:00 A.M. local time and departing 5:00 to 7:00 P.M. local time. Therefore we based the binary measure of temporal boundaries on the time zone difference between cities where members worked (0: during a nine-hour workday, there was at least one hour of overlap, 1: during a nine-hour workday, there were no hours of overlap). For example, there are only four time zones in the continental United States, so all pairs of project members in the United States have overlapping hours in a typical workday. However, for pairs of project members working in the United States and India, the time zone difference is at least 10.5 (West–East or East–West), so they have nonoverlapping hours in a typical workday.

Synchronous Communication. Core members were asked on a 5-point scale (1: Rarely, 2: Monthly, 3: Bi-weekly, 4: Weekly, 5: Daily) "Communication mode is the way that you communicated with fellow members. Please mark how often, during the past six months, you communicated with this person via… (a) voice communication (e.g., telephone), (b) synchronous text communication (e.g., IM), and (c) shared desktop communication (e.g., web conferencing)."

Asynchronous Communication. Core members were asked on a 5-point scale (1: Rarely, 2: Monthly, 3: Bi-weekly, 4: Weekly, 5: Daily) "Communication mode is the way that you communicated with fellow members. Please mark how often, during the past six months, you communicated with this person via... (a) asynchronous text communication (e.g., e-mail)."

For a summary of individual-level and relational-level variables, please see Table 1.

To evaluate the effectiveness of the online survey approach, we calculated reciprocation rates for synchronous and asynchronous communication reported by one member of a pair with communication reported by the other member of the pair. For example, in a project with three members (Persons A, B, and C), we compared the responses of $A \rightarrow B$, $A \rightarrow C$, $B \rightarrow A$, $B \rightarrow C$, $C \rightarrow A$, and $C \rightarrow B$ with the responses of $B \rightarrow A$, $C \rightarrow A$, $A \rightarrow B$, $C \rightarrow B$, $A \rightarrow C$, and $B \rightarrow C$, respectively. We found that 52% of pairs had identical responses, and 79% were within one point on the 5-point scales, which is well within the range of acceptable reciprocation rates (Marsden 1990). We should note that unlike communication frequency that is bidirectional, coordination delay is directional, and should not necessarily have high agreement because Person A could be a source of delay for Person B (but not necessarily the other way around).

Results

The means, standard deviations, and correlations for the pairs of members are found in Table 2. The following two control variables were significantly correlated with the reduction of coordination delay, and remained significant throughout the HLM models: years known ($r = -0.13$, $p < 0.001$) and member interdependence ($r = -0.21$, $p < 0.001$). In support of H1 (see Table 3, Spatial model 1), when pairs of members have overlapping work hours, there was greater coordination delay for those in different cities compared with those in the same city ($B = 0.14$, $p < 0.001$;

Table 1 Summary of Individual-Level and Relational-Level Variables

Level of analysis	Measure	Scale	Range
Individual (level-2)			
1. Project manager	Whether or not respondent was project manager	0: no, 1: yes	0–1
2. Number of members	Number of other members on project	Count	2–32
3. Number of months	Number of months respondent worked on project	Count	1–68
4. Percentage of effort	Percentage of time respondent contributed to project	Percentage of average work week (10% intervals)	1–10
5. Time shifting	Whether or not respondent shifted workday hours	0: no, 1: yes	0–1
Relational (level-1)			
6. Years known	Number of years respondent knew other members	1: < than 1 year; 2: 1–3 years; 3: 3–5 years; 4: 5–10 years; 5: more than 10 years	1–5
7. Member interdependence	(1) Tasks this person performed were related to tasks I performed, (2) this person depended on me for information or materials needed to complete his or her work, (3) I could not accomplish my tasks without information or materials from this person	1: not at all; 3: sometimes; 5: very much	1–5
8. Coordination delay	(1) Typically, it took a long time to get a response from this person, (2) our communications required frequent clarification, (3) we often had to rework tasks beyond what I would normally expect	1: disagree; 3: neutral; 5: agree	1–5
9. Spatial boundaries	Whether or not respondent was in a different city from member	0: same city, 1: different city	0–1
10. Temporal boundaries	Whether or not respondent had overlapping work hours with member	0: at least one hour of overlap, 1: no hours of overlap	0–1
11. Synchronous—Voice	Communication with this person via voice (e.g., telephone) during past six months	1: Rarely, 2: Monthly, 3: Biweekly, 4: Weekly, 5: Daily	1–5
12. Synchronous—Text	Communication with this person via synchronous text (e.g., IM) during past six months	1: Rarely, 2: Monthly, 3: Biweekly, 4: Weekly, 5: Daily	1–5
13. Synchronous—Desktop	Communication with this person via shared desktop (e.g., Web conferencing) during past six months	1: Rarely, 2: Monthly, 3: Biweekly, 4: Weekly, 5: Daily	1–5
14. Asynchronous—E-mail	Communication with this person via e-mail during past six months	1: Rarely, 2: Monthly, 3: Biweekly, 4: Weekly, 5: Daily	1–5

Table 2 Descriptive Statistics and Correlation Matrix for Individual-Level ($N = 675$) and Relational-Level ($N = 5{,}674$) Variables

Variable	N	Mean	SD	1	2	3	4	5	6	7	8	9	10	11	12	13
Individual (level-2)																
1. Project manager	675	0.22	0.42													
2. Number of members	675	12.11	7.06	−0.14												
3. Number of months	675	12.10	12.99	0.09	0.04											
4. Percentage of effort	675	6.17	3.48	0.10	0.13	0.08										
5. Time shifting	675	0.20	0.40	0.06	−0.02	0.02	−0.01									
Relational (level-1)																
6. Years known	5,674	1.79	0.96	0.06	−0.04	0.29	0.00	−0.01								
7. Member interdependence	5,674	2.84	1.35	0.25	−0.20	0.01	0.14	0.00	0.23							
8. Coordination delay	5,674	2.00	0.91	−0.02	0.06	0.02	0.01	0.06	−0.13	−0.21						
9. Spatial boundaries (different city)	5,674	0.70	0.46	−0.03	0.00	0.03	−0.14	0.04	−0.23	−0.21	0.15					
10. Temporal boundaries (no overlapping work hours)	5,674	0.27	0.44	−0.02	0.01	0.04	0.01	0.04	−0.16	−0.13	0.16	0.40				
11. Synchronous—Voice	5,674	2.72	1.47	0.13	−0.25	−0.04	0.10	0.01	0.14	0.58	−0.20	−0.13	−0.10			
12. Synchronous—Text	5,674	1.73	1.29	0.08	−0.18	−0.07	0.09	−0.02	0.07	0.40	−0.13	−0.16	−0.12	0.45		
13. Synchronous—Desktop	5,674	2.12	1.33	0.13	−0.29	−0.03	0.03	−0.07	0.11	0.47	−0.16	−0.04	−0.04	0.60	0.43	
14. Asynchronous—E-mail	5,674	2.96	1.54	0.19	−0.22	0.00	0.20	0.00	0.24	0.73	−0.24	−0.24	−0.12	0.68	0.49	0.58

Note. For $N = 675$, $|r| > 0.10$, $p < 0.01$; $|r| > 0.12$, $p < 0.001$; for $N = 5{,}674$, $|r| > 0.03$, $p < 0.01$; $|r| > 0.04$, $p < 0.001$.

change in model fit statistic, $p < 0.01$). In support of H2 (see Table 3, Temporal model 7), when pairs of members are in different cities, there was greater coordination delay for those with nonoverlapping work hours compared with those who had overlapping work hours ($B = 0.11$, $p < 0.001$; change in model fit statistic, $p < 0.01$).

In support of H3 (see Table 3, Spatial models 2–4), synchronous communication (Voice: $B = -0.04$, $p < 0.001$, Text: $B = -0.04$, $p < 0.001$, Desktop: $B = -0.06$, $p < 0.001$; change in model fit statistics, $p < 0.01$) was negatively associated with coordination delay, and in support of H6 (see Table 3, Temporal model 11), asynchronous communication (E-mail: $B = -0.07$, $p < 0.001$) was negatively associated with coordination delay. And when all communication technology variables are included in the models at once (see Table 3, Spatial model 6 and Temporal model 12), the magnitude of effects decrease somewhat because of the moderate to high correlations among these variables. In examining the HLM model variance components, the relational variables in Table 3 explain 9% of the level-1 variance for Spatial model 6, and 11% of the level-1 variance for Temporal model 12 (Bryk and Raudenbush 1992).

Hypothesis 4, that synchronous communication would reduce coordination delay even more when pairs of members crossed spatial boundaries compared to not crossing spatial boundaries, was not supported (see Table 4, Spatial models 13–15). However, in partial support of H5 (see Table 4, Temporal models 17–19), desktop synchronous communication ($B = 0.04$, $p < 0.05$; change in model fit statistic, $p < 0.05$) reduced coordination delay more when pairs of members had overlapping work hours compared to not having overlapping work hours, but voice synchronous communication and text synchronous communication did not. See Figure 3(a) for an interaction plot, and note that for ease of interpretation, we place synchronous communication on the x-axis (low = 1 standard deviation below the mean, medium = mean, high = 1 standard deviation above the mean) and draw two regression lines based on the binary temporal boundary variable: solid line indicates overlapping work hours and line with squares indicates nonoverlapping work hours. In evaluating

Table 3 **HLM Models Estimating Main Effects on Coordination Delay**

Variable	Spatial (1)	Spatial (2)	Spatial (3)	Spatial (4)	Spatial (5)	Spatial (6)	Temporal (7)	Temporal (8)	Temporal (9)	Temporal (10)	Temporal (11)	Temporal (12)
(Level-2 predictors)												
Project manager	0.08	0.08	0.08	0.08	0.08	0.08	0.14	0.14	0.14	0.14	0.14	0.15
	(0.07)	(0.07)	(0.07)	(0.07)	(0.07)	(0.07)	(0.08)	(0.08)	(0.08)	(0.08)	(0.08)	(0.08)
Number of members	0.01	0.01	0.01	0.01	0.01	0.01	0.00	0.00	0.00	0.00	0.00	0.00
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Number of months	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Percentage of effort	0.01	0.01	0.01	0.01	0.02	0.02	0.01	0.02^{*}	0.02^{*}	0.02^{*}	0.02^{*}	0.02^{*}
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Time shifting	0.10	0.11	0.10	0.10	0.10	0.11	0.09	0.10	0.09	0.08	0.09	0.10
	(0.07)	(0.07)	(0.07)	(0.07)	(0.07)	(0.07)	(0.08)	(0.08)	(0.08)	(0.08)	(0.08)	(0.08)
(Level-1 predictors)												
Years known	-0.06^{***}	-0.06^{***}	-0.06^{***}	-0.05^{***}	-0.04^{***}	-0.04^{***}	-0.07^{***}	-0.07^{***}	-0.07^{***}	-0.07^{***}	-0.06^{***}	-0.06^{***}
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Member interdependence	-0.14^{***}	-0.12^{***}	-0.13^{***}	-0.12^{***}	-0.06^{***}	-0.06^{***}	-0.13^{***}	-0.09^{***}	-0.11^{***}	-0.11^{***}	-0.08^{***}	-0.06^{***}
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Spatial boundaries	0.14^{***}	0.14^{***}	0.14^{***}	0.15^{***}	0.13^{***}	0.13^{***}						
	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)						
Temporal boundaries							0.11^{***}	0.10^{***}	0.10^{***}	0.10^{***}	0.11^{***}	0.10^{***}
							(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)
Synchronous—Voice		-0.04^{***}				0.00		-0.07^{***}				-0.03^{*}
		(0.01)				(0.01)		(0.01)				(0.01)
Synchronous—Text			-0.04^{***}			−0.02			-0.06^{***}			-0.04^{***}
			(0.01)			(0.01)			(0.01)			(0.01)
Synchronous—Desktop				-0.06^{***}		−0.02				-0.06^{***}		-0.04^{**}
				(0.01)		(0.01)				(0.01)		(0.01)
Asynchronous—E-mail					-0.10^{***}	-0.08^{***}					-0.07^{***}	-0.03^{*}
					(0.01)	(0.01)					(0.01)	(0.01)
Model fit	8,213	8,196	8,195	8,189	8,142	8,136	7,587	7,549	7,553	7,542	7,548	7,513
X^2 (Δ Model fit)	34^{***}	17^{***}	18^{***}	24^{***}	71^{***}	77^{***}	22^{***}	38^{***}	34^{***}	45^{***}	39^{***}	74^{***}
(Level-1 df)	(3,615)	(3,614)	(3,614)	(3,614)	(3,614)	(3,611)	(3,292)	(3,391)	(3,391)	(3,391)	(3,391)	(3,206)

Notes. Standard errors are in parentheses below unstandardized coefficients from full maximum likelihood estimation. Spatial models 1–6 include pairs of members *with overlapping work hours*, while Temporal models 7–12 include pairs of members *in different cities*. Degrees of freedom (*df*) for Model fit (i.e., −2 log likelihood) reflect the intercepts not reported here, and the Model fit for a control-only model is 8,247 for Spatial and 7,609 for Temporal. For Level-2 predictors, $df = 541$ in Spatial models and $df = 543$ in Temporal models.

$^{*}p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$.

the simple slopes (Aiken and West 1991), when member pairs have overlapping work hours, the slope is steeply downward and there is a significant negative relationship between synchronous communication and coordination delay (simple slope $= -0.06$, $p < 0.01$). However, when member pairs have nonoverlapping work hours, the slope of the regression line is slightly downward and there is a nonsignificant relationship between synchronous communication and coordination delay (simple slope $= -0.02$, *ns*), indicating that synchronous communication is more valuable for reducing coordination delay when there are overlapping work hours compared to when there are nonoverlapping work hours.

Opposite of what we predicted for H7 (see Table 4, Temporal model 20), e-mail asynchronous communication ($B = 0.06$, $p < 0.001$; change in model fit statistic, $p < 0.01$) reduced coordination delay more when pairs of members had overlapping work hours relative to pairs of members who had nonoverlapping work hours. See Figure 3(b) for an interaction plot, and like Figure 3(a), we place

Table 4 HLM Models Estimating Interaction Effects on Coordination Delay

Variable	Spatial (13)	Spatial (14)	Spatial (15)	Spatial (16)	Temporal (17)	Temporal (18)	Temporal (19)	Temporal (20)
(Level-2 predictors)								
Project manager	0.08	0.08	0.08	0.08	0.15	0.14	0.14	0.15
	(0.08)	(0.08)	(0.08)	(0.08)	(0.08)	(0.08)	(0.08)	(0.08)
Number of members	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Number of months	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Percentage of effort	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Time shifting	0.11	0.11	0.11	0.11	0.10	0.10	0.10	0.09
	(0.07)	(0.07)	(0.07)	(0.07)	(0.08)	(0.08)	(0.08)	(0.08)
(Level-1 predictors)								
Years known	−0.05***	−0.04***	−0.05***	−0.05***	−0.06***	−0.06***	−0.06***	−0.06***
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Member interdependence	−0.06***	−0.06***	−0.06***	−0.06***	−0.06***	−0.06***	−0.06***	−0.06***
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Spatial boundaries	0.13***	0.13***	0.13***	0.13***				
	(0.02)	(0.02)	(0.02)	(0.02)				
Temporal boundaries					0.11***	0.10***	0.11***	0.12***
					(0.02)	(0.02)	(0.02)	(0.02)
Synchronous—Voice	−0.01	−0.01	−0.01	−0.01	−0.04*	−0.03*	−0.03*	−0.03
	(0.01)	(0.02)	(0.01)	(0.01)	(0.02)	(0.02)	(0.01)	(0.01)
Synchronous—Text	−0.02	−0.02	−0.02	−0.02	−0.04***	−0.04**	−0.04**	−0.03**
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Synchronous—Desktop	−0.01	−0.01	−0.03	−0.01	−0.04**	−0.04**	−0.06***	−0.04**
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Asynchronous—E-mail	−0.09***	−0.09***	−0.08***	−0.09***	−0.03*	−0.03*	−0.03*	−0.05***
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Spatial/temporal ∗ Voice	0.02				0.02			
	(0.01)				(0.01)			
Spatial/temporal ∗ Text		0.00				0.01		
		(0.01)				(0.02)		
Spatial/temporal ∗ Desktop			0.02				0.04*	
			(0.01)				(0.02)	
Spatial/temporal ∗ E-mail				0.02				0.05***
				(0.01)				(0.01)
Model fit	8,136	8,136	8,136	8,135	7,510	7,512	7,506	7,500
X^2 (Δ Model fit)	0	0	0	1	3	1	7*	13***
(Level-1 df)	(3,610)	(3,610)	(3,610)	(3,610)	(3,387)	(3,387)	(3,387)	(3,387)

Notes. Standard errors are in parentheses below unstandardized coefficients from full maximum likelihood estimation. Spatial models 13–16 include pairs of members *with overlapping work hours*. Temporal models 17–20 include pairs of members *in different cities*. Degrees of freedom (df) for Model fit (i.e., -2 log likelihood) reflect the intercepts not reported here. For level-2 predictors, $df = 541$ in Spatial models and $df = 543$ in Temporal models.
$^*p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$.

asynchronous communication on the x-axis (low = 1 standard deviation below the mean, medium = mean, high = 1 standard deviation above the mean) and draw two regression lines based on the binary temporal boundary variable: solid line indicates overlapping work hours and line with squares indicates nonoverlapping work hours. When the members have overlapping work hours, the slope is downward and there is a significant negative relationship between asynchronous communication and

Figure 3(a) **Interaction Plot of Synchronous Communication and Temporal Boundaries (i.e., Nonoverlapping Work Hours) on Coordination Delay in Globally Distributed Projects**

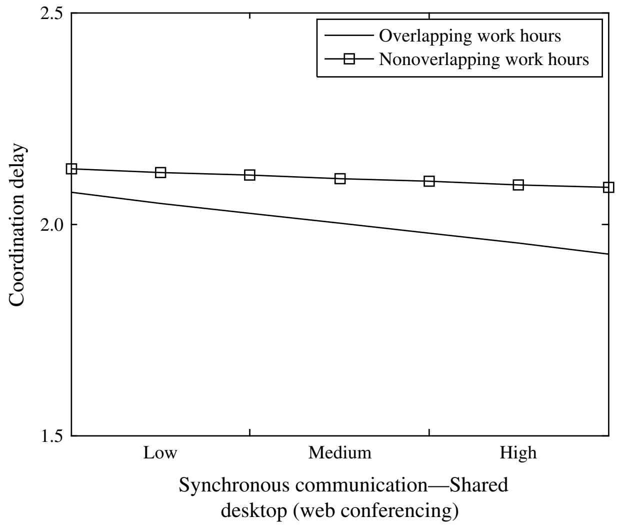
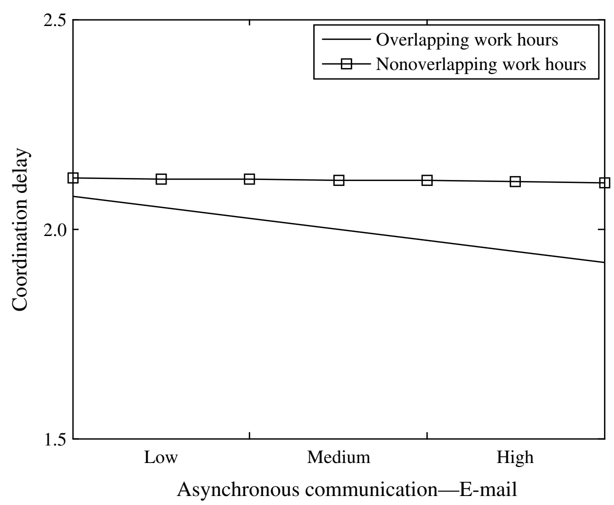


Figure 3(b) **Interaction Plot of Asynchronous Communication and Temporal Boundaries (i.e., Nonoverlapping Work Hours) on Coordination Delay in Globally Distributed Projects**



coordination delay (simple slope = −0.06, $p < 0.01$). However, when the members have nonoverlapping work hours, the slope of the regression line is slightly downward and there is a nonsignificant relationship between asynchronous communication and coordination delay (simple slope = −0.01, *ns*), indicating that asynchronous communication is more valuable for reducing coordination delay when there are overlapping work hours compared to when there are nonoverlapping work hours, again opposite of what we had predicted.

Finally, in exploratory analyses, we examined whether coordination delay was associated with project performance by averaging coordination delay at the project level ($N = 108$ projects). As part of the online survey, project managers, and members were asked to complete a three-item performance scale: "Overall, to what extent do you disagree or agree with the following… we completed the work on schedule/on time, we completed the work well within budget, the final product met requirements" (1: disagree; 3: neutral; 5: agree; Cronbach's $\alpha = 0.81$; intraclass correlation = 0.19, $p < 0.01$) (adapted from Ancona and Caldwell 1992). At the project level, when coordination delay was entered into a model with control variables, it was significantly negatively associated with project performance ($B = -0.43$, $p < 0.01$, $\Delta R^2 = 0.07$, $F_{\text{change}} = 7.52$, $p < 0.01$). Note that even after controlling for the same variables used in the HLM models, we did not find a direct relationship between spatial or temporal boundaries and project performance. Furthermore, none of the communication technology variables reduced the negative association between coordination delay and project performance, and neither boundary moderated the relationship between coordination delay and project performance. These results reinforce the importance of understanding coordination delay among pairs of members when spatial and temporal boundaries exist, thus highlighting the value of our relational model of coordination delay.

Discussion

We contribute to the growing literature on distributed work, and extend coordination theory by conceptually and empirically distinguishing between the impact of spatial and temporal boundaries on coordination delay in globally distributed projects. To the best of our knowledge, this is the first field study to systematically evaluate differences among spatial and temporal boundaries, and more important, to examine how these differences affect the success of communication technologies in helping to overcome coordination delay. While there is a long history of research on the role of communication technology in facilitating distributed work, literatures ranging from computer-supported cooperative work to engineering management to information systems have primarily focused on spatial boundaries (Allen 1977, O'Leary and Cummings 2007, Olson and Olson 2000), without controlling for or explicitly considering temporal boundaries. Furthermore, when scholars have examined the influence of time and timing in distributed projects, the focus has been on the sequencing or progression of activities (e.g., Massey et al. 2003) rather than the temporal boundaries that influence when project members have time available to work.

Methodologically, our study also makes an important advance by testing a relational model of coordination delay that addresses both spatial boundaries (same city versus different city with overlapping work hours) and temporal boundaries (overlapping work hours versus nonoverlapping work hours in different cities). Focusing on pairs of members in globally distributed projects provides insights that aggregating to the project level does not allow. Much of the work in globally distributed projects is done individually or with a few other members. Rarely does an entire project team work on the same task at the same time. Therefore, by disaggregating the project into pairs of members, we can better understand how spatial and temporal boundaries predict coordination delay within the project. Not only does this approach allow us to identify the source of coordination delay with greater accuracy, but it also allows managers to target potential solutions to specific pairs of members (e.g., time shifting, work partitioning, or even collocation). This is why studying pairs of members in globally distributed projects, rather than simply taking a project average of coordination delay, is essential for advancing research in this area.

We also believe that by focusing on a relational model of coordination delay, we are tapping into the practical problems faced by pairs of globally distributed project members. In support of the hypotheses, we empirically demonstrate that both spatial and temporal boundaries are associated with coordination delays, and that synchronous (i.e., web conferencing) and asynchronous communication (i.e., e-mail) help to reduce coordination delays for pairs of members who *share overlapping work hours*. This has important implications for managers because it can help them make spatial and temporal configuration decisions for where members are in projects and allows them to deploy the most effective communication technology for that configuration. Our results suggest that simply having a small amount of overlap in work hours among sites can help members take greater advantage of communication technologies. While we are not aware of prior research that has linked the spatial and temporal configuration of project members to communication technology use, the following comment provided by a globally distributed project member we interviewed supports our results:

> One [communication technology] that is useful in the same time zone or in nearly the same time zone to where you're both at work together is web conferencing. That one is very useful because you can see what's going on and you can have ten people debugging a problem and looking at the same code at the same time and we don't have to be there at the same place. Now it is also useful for other people in different time zones, but it is not useful if you're not awake at the same time and working at the same time. So, web conferencing is a very good tool if you've got people that are working together just in different geographies, but they are still working and awake.

However, we also found that for pairs of members who do *not* share overlapping work hours, the use of communication technology does not provide an advantage. Thus, temporal boundaries appear to be more difficult to cross with communication technologies than spatial boundaries. This also has important implications for practice because it makes evident the urgency to identify other types of collaboration strategies to help projects reduce coordination delay. Coordination theory suggests that mechanistic forms of coordination such as routines, procedures, and collaboration tools are more effective than direct communication when the task is more predictable and routine (March and Simon 1958), but it may be possible that these forms of coordination are useful or even necessary to make up for the lack of direct communication when there are temporal boundaries. A globally distributed project member we interviewed described the challenge of being separated by temporal boundaries:

> It's a day turn around. It always is. Unless, for example, last night I had to ask a question. It was about 11:00 last night and I found somebody that was online on

> Instant Messenger and I was able to communicate with them that way. So those tools help, but typically it is a day turnaround because the people that we're working with are on the other side of the globe and so they are up when we're asleep and vice versa.

Another important implication of our study is that through exploratory analyses, we established that spatial and temporal boundaries do not directly affect project performance, but rather they do so indirectly through relational processes like coordination delay. It was not the case that taking longer to coordinate work led to a higher-quality solution, rather it inhibited progress on the task. This insight can help managers identify a path linking spatial and temporal configuration of members to project performance. By analyzing the spatial and temporal boundaries, as well as the communication technologies used within each relationship on a project, managers can intervene to help reduce coordination delay for each pair of members. In situations where managers have little leeway in changing the spatial and temporal configuration of members, our findings can help them evaluate the trade-offs of trying to encourage direct communication versus other noncommunication alternatives given the spatial and temporal boundaries of members.

A rising trend in the area of software development is the use of agile methodologies and other approaches that encourage project members to communicate informally, without the need for plans and fixed requirements (Ramesh et al. 2006). We view coordination delay as a major hindrance to the effective application of these kinds of methodologies. That is, coordination delay decreases the ability of project members to be flexible and agile in their work. An interesting question raised by the current research is whether temporal boundaries pose a greater challenge for agility than spatial boundaries. For projects that are radically colocated, and have opportunities to meet face to face to immediately resolve problems that arise, these kinds of new methodologies seem promising (Teasley et al. 2002). For geographically dispersed projects, particularly those spanning large time zone differences, it is difficult to ensure that processes will be in place to support collaboration when members are not available for synchronous communication. However, our study suggests that if project members are spread out across multiple time zones, but have some work time overlap, they may be in a better position to use agile methods and make effective use of communication tools. Recent work by Sarker and Sarker (2009) supports this implication, and they suggest that tactics such as taking into account time zones when scheduling (short and frequent) meetings in addition to temporarily staggering work time may help project members bridge time zones through synchronous communication.

Our results also suggest that prior experience is generally helpful for reducing coordination delay. Thus, any methodologies that encourage IT or engineering personnel to work for a short period of time with others they have not met before may be problematic in globally distributed projects. We believe it is critical to pay attention to the shared experiences that members have had in prior projects rather than rapidly moving people on and off projects. In our study, we also find that member interdependence is helpful for reducing coordination delay. It may be that engaging in noncommunication activities to coordinate work, such as preestablished schedules, division of labor, and work routines may be an alternative to communication technology use (Olson and Olson 2000). Again, for methodologies that tend to minimize formal planning, members of globally distributed projects may miss out on opportunities to effectively manage dependencies by thinking through the coordination needs in advance of the work.

Previous research has also illustrated how developing common practices for dispersed coordination is difficult and requires aligning the effort of all parties involved (Orlikowski 2002). Our interviews suggest there may be value in encouraging division of responsibilities across sites, such that members in different locations have schedules and work routines that minimize the need for overlap (and hence the potential for coordination delay). Along these lines, a globally distributed project member said:

> One of the problems that we have when working with people that are eight hours away in time zones, is the coordination of large projects... the engineers need to work together to talk through problems. So, when there are significant time differences, they just can't make good solid progress without being able to talk. So, what we started to do in the last couple of years is starting to divide up the project to where each site has

> a completely separate job, with some minor overlap with the other sites. But in general, they have one complete section of the project all to their own. Otherwise, what we found in trying to manage one functionality across multiple sites is that people were having to work a ridiculous number of hours in order to maintain that communication.

Limitations and Future Directions

In exploring globally distributed projects in a single organization, we limit the generalizability of our results to technical projects with geographically dispersed members working in large multinational organizations. Smaller companies, or firms with only a few geographic locations, or even those with different corporate cultures, may face other issues not described in this study. We also realize that spatial distance could be conceptualized as the number of miles between project members, and temporal distance could be conceptualized as the number of time zones between project members. However, in our data set, the number of miles and number of time zones within pairs of members were correlated $\underline{r} = 0.95$, making a comparison of this alternative conceptualization of distance infeasible. We also did not evaluate the configuration of members within and across spatial and temporal boundaries (e.g., three members in the United States, two members in the United Kingdom, one member in India), which could affect coordination delay (O'Leary and Cummings 2007). We encourage other researchers to look for additional ways to further tease apart the measurement and configuration of spatial and temporal boundaries, for example, by examining projects with the same number of members in North and South America, so that there are spatial boundaries but not temporal boundaries, while keeping the configuration balanced. In our sample, the configurations varied quite a bit and pairs of project members were in the same country separated North–South or were in different countries separated East–West.

We also find the issue of time shifting very interesting, even though in our study, only about 20% of respondents reported working outside of a typical workday (and it was *not* significantly associated with coordination delay in any of the HLM models, nor did it moderate the relationship between temporal boundaries and coordination delay). In-depth qualitative analyses and field interviews may shed more light on the advantages and disadvantages of working during the middle of the night, or shifting hours in the workday to better overlap with project members in other geographic locations. Several of the globally distributed project members we interviewed felt strongly about how time shifting helped them reduce coordination delay, including someone who said:

> I think one thing that's been great, and I haven't seen it as much in other [projects], but we have all shifted our hours. I shift, so the rest of the team has shifted their hours where they will work 11:00 A.M. to 8:00 P.M. their time. I work 4:00 A.M. to noon or so. What happens is, is that gives us a six-hour overlap or more. So we overlap our time. Israel shifts their hours somewhat so that they overlap more with Russia and maybe an hour or two with the U.S. So everybody's sacrificing a little bit so that we can have as much time as we're allowed together that we can. I've seen it the other way where you just have to use daily handoffs. Those are much harder to make sure that you're all on track and all hearing the same message. At least one-hour meetings that we're all there, things happen much quicker.

There may also be cultural differences in how project members in different countries use their time as well as their technology. For example, members in the United States and India may differ in norms of what is acceptable communication outside of typical business hours. We conducted exploratory analyses on this issue, although country differences (a proxy for cultural differences) did not affect our results on coordination delay or communication technology use. There is also the issue of transportation time, because in Europe, it takes much less time to travel from one country to another than it does to travel from the United States to a country in Europe. In some parts of the world, members can be in different cities, but have more opportunities to hold face-to-face meetings and discussions at critical points in the project life cycle (for example, at the beginning or middle of a project).

Along with investigating differences between spatial and temporal boundaries, and how time shifting affects globally distributed project effectiveness, there are a number of other theoretically rich avenues for exploring "virtuality." Following the lead of Griffith et al. (2003) and Kirkman and Mathieu (2005), we need to learn more about the extent to which project members are supported in their use of communication technology, as well as how members are supported when they are apart from other members. For example, project managers who travel a lot may have access to different levels of technology (e.g., broadband Internet access versus dial-up or mobile Internet access). Similarly, the technical support for communication tools may be greater in some regions of the world than others, depending on the number of employees at a particular site or the resources available to employees. There are certainly an increasing variety of technologies available to help project members communicate across space and time, although it may take awhile for them to achieve the critical mass of the telephone and e-mail. And finally, we have not examined other factors that prior research has identified as being important for distributed work, such as changes in technology use over time (Majchrzak et al. 2000), general levels of trust (Jarvenpaa and Leidner 1999), and other forms of diversity in globally distributed projects (Cummings 2004).

Conclusion

There are many ways to manage a globally distributed project. Our results suggest that the use of communication technology with project members who are spread across spatial and temporal boundaries does not eliminate the problem of coordination delay. For members working across spatial and temporal boundaries, relying simply on e-mail can be a struggle. As one member of a globally distributed project stated in an interview:

> For example with the e-mails, after a couple of e-mails you can tell that it is difficult to communicate through the e-mails and my point might not be getting across or their point is not getting across so it is a lot easier just to make the phone call after a couple of e-mails. So sometimes it is a problem because you'll see that there are multiple e-mails that are going on and it just keeps on going. And there are different people within the team that are asking these questions and so I have to put a stop to it and basically call a meeting to get it resolved because there is just no way; the e-mails are not specific enough and it is not a good enough communication tool so people can completely understand what the problem is, what are the things that they are facing and what exactly do they need from us or what we need from them.

Given that the impact of spatial and temporal boundaries on coordination delay does not disappear with the use of communication technologies such as e-mail, we suggest that managers might consider one of several possible solutions: (1) work redesign, (2) tool enhancement, or (3) job training. First, the work redesign solution includes assigning to globally distributed projects members who have at least some overlapping hours in their workday. For example, in the case of a project that spans the United States, and India, having a member in the United Kingdom could help in coordinating the workflow. When possible, managers could look for opportunities for members to work together longitudinally (rather than latitudinally), such as assigning members who work together North–South. When changing the geographic configuration of the project is less feasible, managers can strive to allocate tasks with greater dependencies to members who have more work time overlap, and thus are at less risk for coordination delay.

Second, the tool enhancement solution includes next generation collaboration applications that could help projects coordinate their work without relying so heavily on direct real-time communication. For example, software developers could explore building tools that display the spatial and temporal profiles of project members to highlight the separation of member pairs (in both physical location and time zone). The visualization of spatial and temporal boundaries, coupled with details about workflow and member availability, could facilitate coordination. Finally, the job training solution includes helping project members to better partition work (e.g., dividing up responsibilities), plan for dependencies in the tasks, and synchronize the hand off of individual pieces. This should help to facilitate work when members do not have overlapping hours, but need to be involved in the project together. In short, work redesign, tool enhancement, and job training solutions that help with task organization, rather than simply task communication, could enable globally distributed projects affected by spatial and temporal boundaries to reduce coordination delays.

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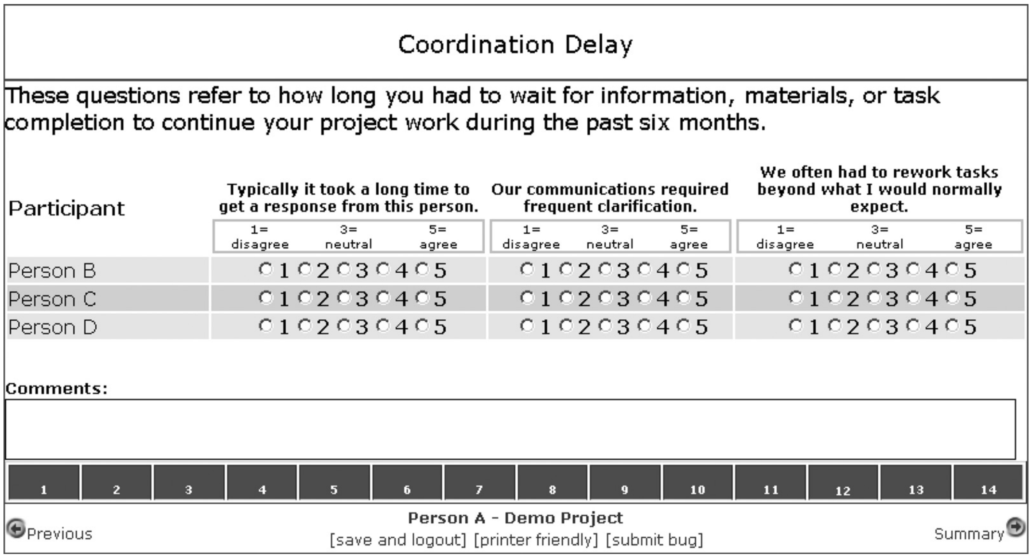
Appendix. An Illustration of the Data Collection Method Used to Test a Relational Model of Coordination Delay in Globally Distributed Projects

Because the main focus of our research is on member pairs (rather than the project as a whole), it is important to highlight how we measured relational variables such as coordination delay. In the below screen capture of our online survey, we display the webpage seen by Person A (a member of the "Demo Project"), who is responding to questions about coordination delay for Person B, Person C, and Person D. In contrast to measures where each respondent judges the overall amount of coordination delay in the project, relational measures require each respondent to assess the amount of coordination delay with every other project member.



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